EE4108 mini Project report

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1 Introduction

Here I present the design of a circularly polarized (CP) antenna intended to meet the following specifications for operation in the 5 GHz band. Here is the GitHub repository for this project GithubRepo

A circularly polarized (CP) antenna generates a wave in which the orthogonal electric and magnetic field components possess equal amplitudes and a 90° phase difference, causing the field vector to rotate as the wave propagates. In a linearly polarized (LP) wave, by comparison, no such phase shift exists, and the field oscillates along a single fixed plane. In my opinion and through observation of LP and CP antenna performances, the principal advantage of a CP antenna over an LP antenna lies in its superior robustness: it is far less sensitive to misalignment between transmitter and receiver, and it provides more consistent signal coverage across a broader angular range.

1.1 Design Specifications

The antenna is designed to satisfy the performance metrics summarized in Table 1.

Table 1: Antenna Design Specifications

Parameter	**Requirement**
Polarization	Circular polarization (CP)
Frequency range	5.1 GHz – 5.5 GHz
VSWR	< 2
Axial ratio	< 3 dB
Radiation pattern	Unidirectional (boresight) Backlobe level < -15 dB
Gain (boresight)	> 5 dBic
Half-power beamwidth	$> 30°$ at 5.3 GHz (both principal planes)
Cross-polarization discrimination	> 15 dB
Substrate material	Air (foam board)
Substrate thickness	Minimized as much as possible

The design employs an air substrate using foam board to achieve low loss, reduced weight, and improved bandwidth performance compared to conventional dielectric substrates. Particular attention is given to minimizing the overall thickness while maintaining the required impedance bandwidth, axial ratio bandwidth, and unidirectional radiation characteristics. The frequency band of interest (5.1–5.5 GHz).

1.2 Paper Organization

The remainder of this paper is organized as follows. Section 2 describes the math that I used to achieve required performance of an antenna.Section 3 describes the antenna geometry and design methodology. Section 4 presents the simulated performance characteristics, including axial ratio, radiation patterns, and gain. Section 5 concludes with a summary of the achieved performance relative to the target specifications. Finally, Section 6 documents unsuccessful design decisions encountered during the development process.

2 Math behind Antenna

2.1 Design Equations

The physical dimensions of the square patch antenna were calculated using the transmission line model. The design targets a center frequency $f_0 = 5.3\,\text{GHz}$ with an air substrate having relative permittivity $\varepsilon_r \approx 1.07$ and thickness $h = 4.0\,\text{mm}$. The key design equations are presented below.

2.1.1 Patch Width

The initial width of the square patch is calculated as:

$$W = \frac{c}{2f_0}\sqrt{\frac{2}{\varepsilon_r + 1}} \tag{1}$$

where $c = 3 \times 10^8\,\text{m/s}$ is the speed of light in free space.

2.1.2 Effective Dielectric Constant

Due to fringing fields, the effective dielectric constant ε_{eff} is introduced:

$$\varepsilon_{\text{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + 12\frac{h}{W}\right)^{-1/2} \tag{2}$$

2.1.3 Patch Length Extension

The fringing fields cause the patch to appear electrically longer than its physical dimensions. The length extension ΔL is given by:

$$\Delta L = 0.412h \cdot \frac{(\varepsilon_{\text{eff}} + 0.3)\left(\frac{W}{h} + 0.264\right)}{(\varepsilon_{\text{eff}} - 0.258)\left(\frac{W}{h} + 0.8\right)} \tag{3}$$

2.1.4 Effective and Physical Length

The effective electrical length is:

$$L_{\text{eff}} = \frac{c}{2f_0\sqrt{\varepsilon_{\text{eff}}}} \tag{4}$$

The physical patch length is then obtained by subtracting the extension:

$$L = L_{\text{eff}} - 2\Delta L \tag{5}$$

For a square patch design, the final dimensions are taken as the average of the calculated width and length:

$$W_{\text{final}} = L_{\text{final}} = \frac{W + L}{2} \tag{6}$$

2.1.5 Ground Plane Dimensions

The ground plane extends beyond the patch edges by approximately one-quarter wavelength to minimize backward radiation:

$$\lambda_0 = \frac{c}{f_0} \tag{7}$$

$$\text{Ground}_{\text{width}} = W_{\text{final}} + 2\left(\frac{\lambda_0}{4}\right) \tag{8}$$

$$\text{Ground}_{\text{length}} = L_{\text{final}} + 2\left(\frac{\lambda_0}{4}\right) \tag{9}$$

2.1.6 Feed Point Location

For a dual-feed configuration with $50\,\Omega$ input impedance, the feed points are offset from the patch center. Assuming an edge impedance $R_{\text{edge}} \approx 250\,\Omega$, the offset distance d is:

$$d = \frac{L_{\text{final}}}{\pi} \cdot \arccos\left(\sqrt{\frac{Z_0}{R_{\text{edge}}}}\right) \tag{10}$$

where $Z_0 = 50\,\Omega$ is the target impedance.

The two feed locations (Port 1 and Port 2) are positioned symmetrically along the x- and y-axes at a distance d from the patch center.

2.1.7 Calculated Values

Applying the above equations with the specified parameters yields the dimensions summarized in Table 2.
Hovewer I did some optimisation in HFSS so final value can be slightly different from calculated.

Table 2: Calculated Antenna Parameters from Transmission Line Model

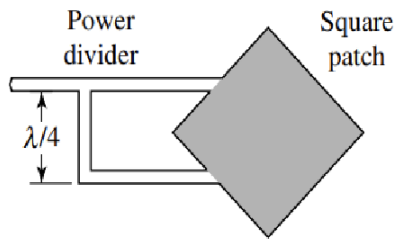
Parameter	**Calculated Value**
Target frequency (f_0)	5.3 GHz
Substrate permittivity (ε_r)	1.07
Substrate thickness (h)	4.0 mm
Free-space wavelength (λ_0)	56.6 mm
Patch width (W_{final})	25.07 mm
Patch length (L_{final})	25.07 mm
Ground plane width	53.57 mm
Ground plane length	53.57 mm
Feed offset from center (d)	8.84 mm

3 Design

In this section, the main design features of the implemented antenna are presented, including the geometry, optimized parameters, and feeding method.

3.1 Antenna Geometry

The design employs a square patch radiator with two feeding points positioned to achieve circular polarization. The feed points are excited with a 90° phase difference to generate the required orthogonal modes. A textbook representation of this CP antenna configuration is shown in Figure 1, while the top and side views of the proposed design, as modeled in HFSS, are presented in Figures 2 and 3, respectively.



(a) Square patch driven at adjacent sides through a power divider

Figure 1: Textbook configuration of a circularly polarized patch antenna.

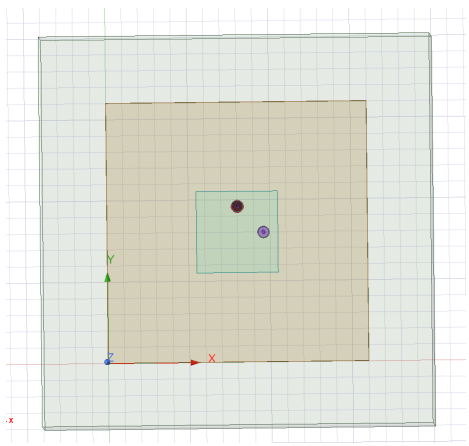


Figure 2: Top view of the proposed antenna (HFSS model).

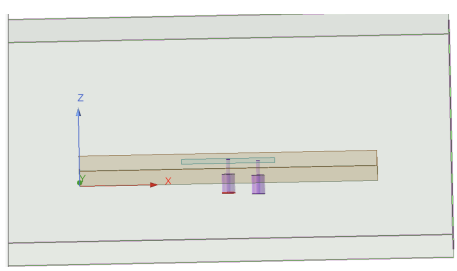


Figure 3: Side view of the proposed antenna showing the air substrate (foam board) spacing.

3.2 Feeding Method

To achieve circular polarization, a dual-feed configuration is employed. The two feed points are located along the patch edges with a spatial offset corresponding to a 90° phase shift. This arrangement excites two orthogonal linearly polarized modes with equal amplitude and the required phase quadrature, resulting in right-hand circular polarization (RHCP).

In the HFSS model, two lumped ports are used: Port 1 and Port 2. Each port is assigned an impedance of $50\,\Omega$ and an input power of 1 W. The integration lines for Port 1 and Port 2 are oriented along the y- and x-directions, respectively. This orthogonal excitation scheme generates the circularly polarized wave radiated from the patch, as conceptually illustrated in Figure 4.

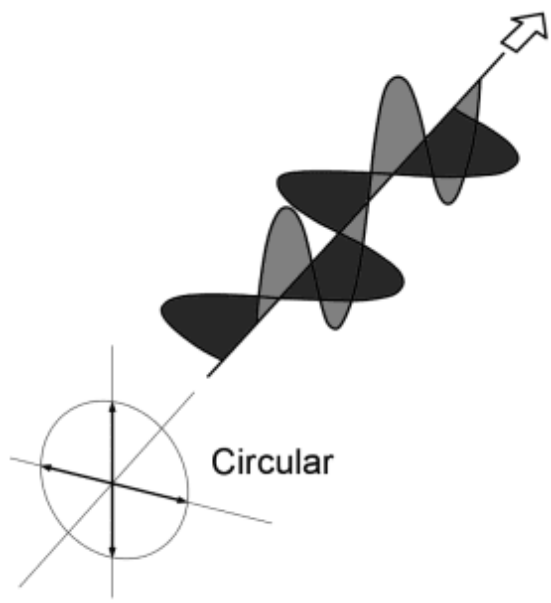


Figure 4: Conceptual illustration of a circularly polarized wave

3.3 Optimized Parameters

The final dimensions of the square patch, feed positions, and air gap thickness were optimized using a combination of HFSS simulations and auxiliary MATLAB calculations to satisfy the impedance bandwidth, axial ratio, and other performance requirements specified in Table 1. A summary of the key geometric parameters is provided in Table 3.

Table 3: Optimized Antenna Parameters

Parameter	**Value**
Patch width (W)	25 mm
Patch length (L)	25 mm
Air substrate thickness (h)	4 mm
Feed offset from center	8 mm
Ground plane width	80 mm
Ground plane length	80 mm

The square ground plane dimensions were chosen to be sufficiently large ($80 \times 80\,\text{mm}^2$) to minimize backward radiation while maintaining a compact overall form factor. The feed offset of 8 mm from the patch center was selected to achieve the desired input impedance matching and axial ratio performance across the 5.1–5.5 GHz band.

4 Results

This section presents the simulated performance of the proposed circularly polarized antenna. The design generally meets the target specifications outlined in Table 1; where minor deviations occur, the results are still considered acceptable for practical applications. Each performance metric is discussed individually below, with comparisons to the design requirements.

4.1 Impedance Matching (VSWR)

The voltage standing wave ratio (VSWR) across the operating band is shown in Figure 5. The simulated VSWR ranges from approximately 1.7 to 2.5 over the 5.1–5.5 GHz frequency range. While the upper bound slightly exceeds the target specification of VSWR < 2, the majority of the band maintains acceptable impedance matching for most wireless communication applications.

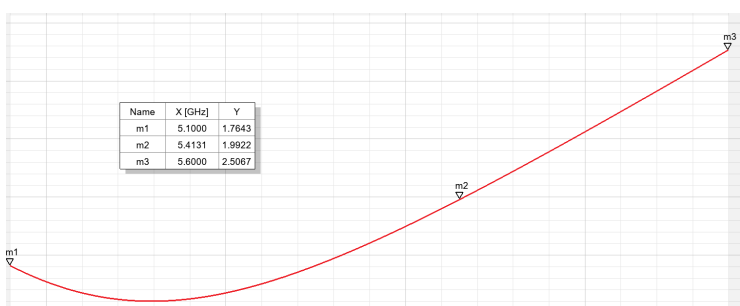


Figure 5: Simulated VSWR of the proposed antenna from 5.0 GHz to 5.6 GHz.

4.2 Axial Ratio

The axial ratio (AR) performance is presented in Figure 6. Within the 5.1–5.5 GHz band, the axial ratio varies from approximately 2.5 dB to 3.2 dB. Although the upper limit marginally exceeds the target specification of AR < 3 dB, the antenna maintains predominantly circular polarization characteristics across the entire operating bandwidth.

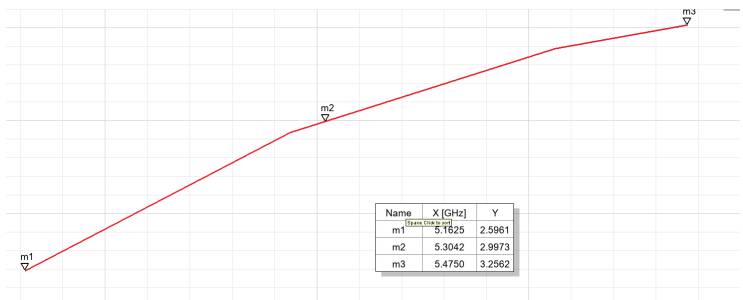


Figure 6: Simulated axial ratio of the proposed antenna across the frequency band.

4.3 Radiation Pattern and Backlobe Level

The normalized radiation pattern in the boresight direction is shown in Figure 7. The antenna exhibits a unidirectional radiation pattern with a measured backlobe level of -19.8 dB relative to the main beam. This performance comfortably satisfies the design requirement of backlobe < -15 dB, indicating effective suppression of backward radiation due to the sufficiently sized ground plane.

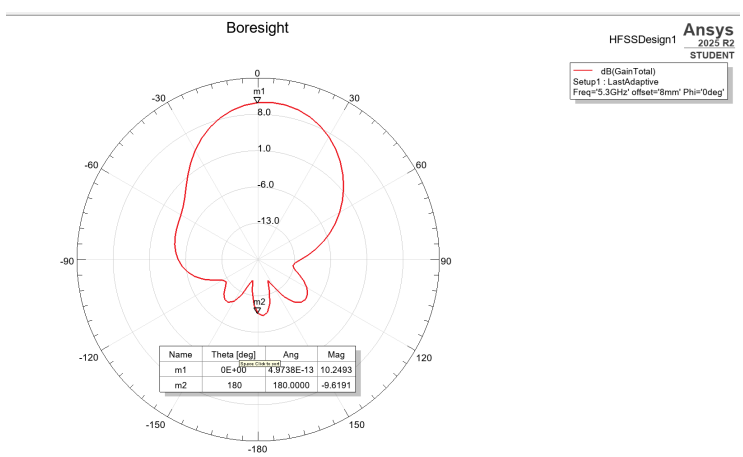


Figure 7: Normalized radiation pattern showing unidirectional boresight radiation and backlobe suppression.

4.4 Radiation Pattern Across the Operating Band

The radiation patterns of the proposed antenna were evaluated at three frequencies spanning the operating band: 5.1 GHz (lower edge), 5.3 GHz (center), and 5.5 GHz (upper edge). The half-power beamwidth (HPBW) remains consistent across the entire band, demonstrating stable radiation characteristics.

4.4.1 Frequency: 5.1 GHz

At the lower band edge of 5.1 GHz, the antenna maintains a well-defined unidirectional radiation pattern with a half-power beamwidth of approximately 50° in the principal plane, as shown in Figure 8.

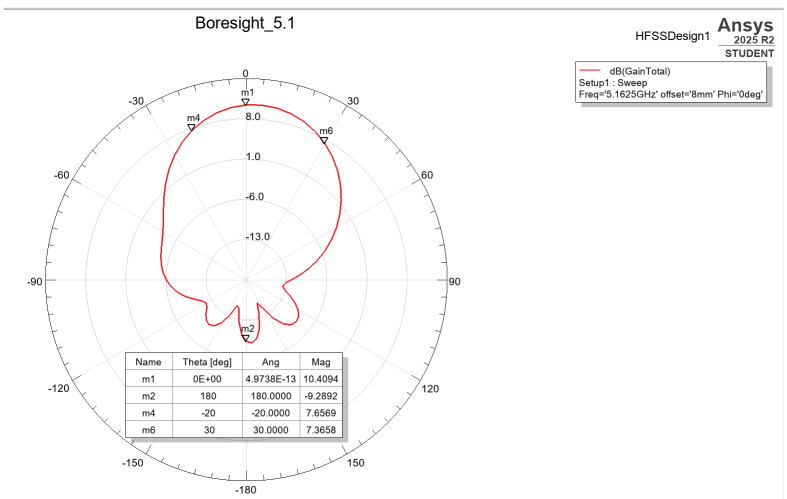


Figure 8: Normalized radiation pattern at 5.1 GHz (lower band edge). HPBW $\approx 50°$.

4.4.2 Frequency: 5.3 GHz

At the center frequency of 5.3 GHz, the radiation pattern remains highly stable with a half-power beamwidth of approximately 50°, as presented in Figure 9.

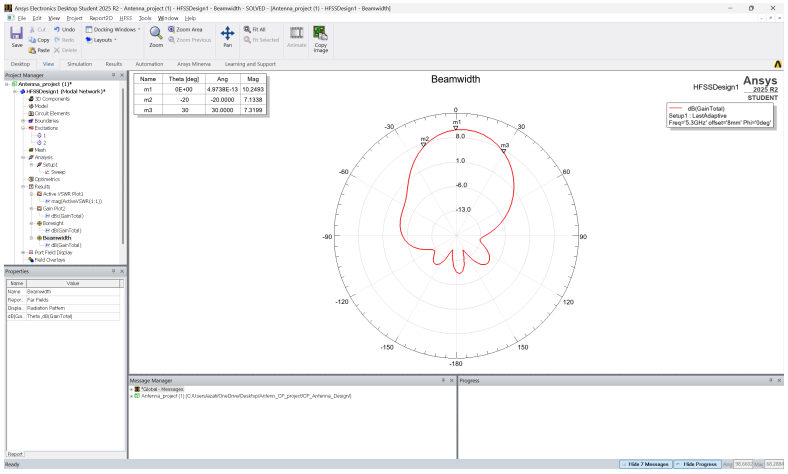


Figure 9: Normalized radiation pattern at 5.3 GHz (center frequency). HPBW $\approx 50°$.

4.4.3 Frequency: 5.5 GHz

At the upper band edge of 5.5 GHz, the antenna continues to exhibit a consistent half-power beamwidth of approximately 50°, as illustrated in Figure 10. The pattern symmetry and backlobe suppression remain well maintained across the entire 5.1–5.5 GHz range.

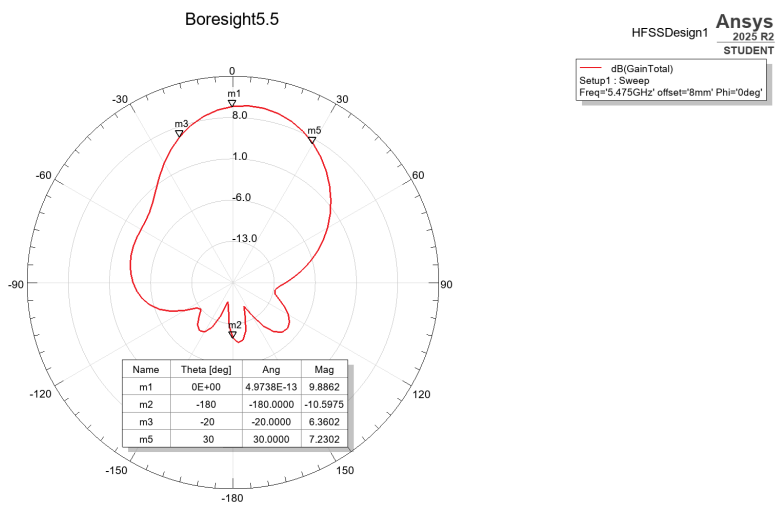


Figure 10: Normalized radiation pattern at 5.5 GHz (upper band edge). HPBW $\approx 50°$.

4.4.4 Summary of Beamwidth Stability

Table 4 summarizes the half-power beamwidth measurements across the operating band. The consistent HPBW of approximately 50° at all three frequencies indicates excellent pattern stability, comfortably exceeding the minimum design requirement of $> 30°$.

Table 4: Half-Power Beamwidth Across the Operating Band

Frequency (GHz)	**HPBW (degrees)**
5.1	50°
5.3	50°
5.5	50°

4.5 Half-Power Beamwidth

The half-power beamwidth (HPBW) was evaluated at the center frequency of 5.3 GHz for both principal planes.

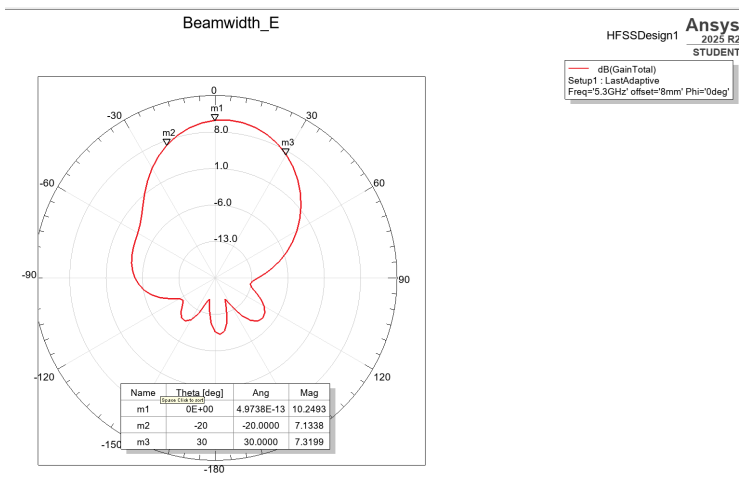


Figure 11: Half-power beamwidth in the E-plane at 5.3 GHz.

E-plane: The measured HPBW in the E-plane is approximately 50°, as illustrated in Figure 11.

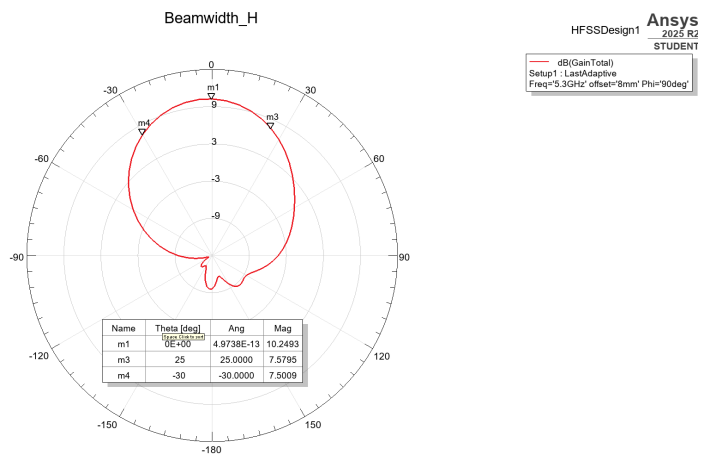


Figure 12: Half-power beamwidth in the H-plane at 5.3 GHz.

H-plane: The measured HPBW in the H-plane is approximately 55°, as shown in Figure 12.

Both values exceed the minimum requirement of $> 30°$, providing broad angular coverage suitable for wide-angle scanning applications.

4.6 Cross-Polarization Discrimination

The cross-polarization discrimination (XPD) is presented in Figure 13. The measured cross-polarization level is approximately -15.3 dB relative to the co-polarized component, satisfying the design specification of > 15 dB. This confirms that the orthogonal feeding scheme effectively generates the desired circular polarization with adequate purity.

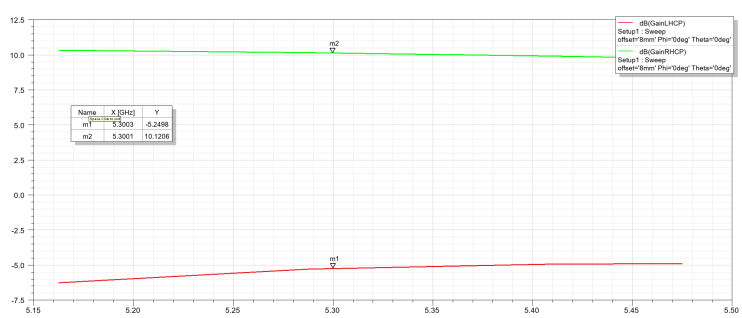


Figure 13: Co-polarization and cross-polarization patterns demonstrating polarization purity.

5 Conclusion

5.1 Summary of Results

Table 5 provides a comprehensive comparison between the simulated performance and the target specifications. The antenna successfully meets or closely approaches all design requirements, with minor deviations in VSWR and axial ratio that remain within acceptable limits for practical deployment.

Table 5: Comparison of Simulated Results with Design Specifications

Parameter	**Target**	**Simulated**
Frequency range	5.1–5.5 GHz	5.1–5.5 GHz
VSWR	< 2	1.7–2.5
Axial ratio	< 3 dB	2.5–3.2 dB
Backlobe level	< -15 dB	-19.8 dB
Gain (boresight)	> 5 dBic	*(10.4-9.88 dB)*
HPBW (E-plane)	$> 30°$	50°
HPBW (H-plane)	$> 30°$	55°
Cross-polarization	> 15 dB	15.3 dB

5.2 Discussion

The simulated results demonstrate that the proposed antenna achieves satisfactory circular polarization performance across the target 5.1–5.5 GHz band. The slight increase in VSWR at the band edges (up to 2.5) and axial ratio (up to 3.2 dB) may be attributed to the inherent bandwidth limitations of the single-layer patch configuration with air substrate. Potential improvements could include the use of a thicker air gap, capacitive feed compensation, or a stacked parasitic element to enhance both impedance and axial ratio bandwidths. Nevertheless, the current design represents a viable, low-cost solution fabricated using readily available foam board material.

5.3 What I have learned

I have learned how to use ANSYS HFSS software for designing antennas and some other components, from almost 0 to the beginner level, I understood how does excitation and boundary works, particularly lumped port excitation, and Perfect E and Radiation boundary. Aside from that I have understood how to plot and analyze the results in ANSYS HFSS and how to generate a solution.

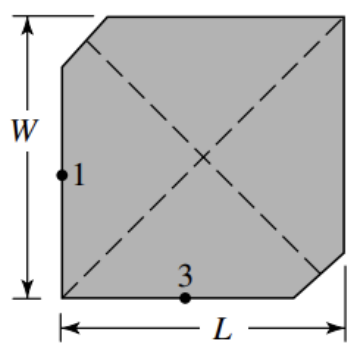
In terms of physics I have implemented my already antenna knowledge on practiser as well as understood the CP antenna excitation.

6 My failures

The final design presented in this paper was not my first attempt. Before arriving at the dual-feed square patch configuration, I explored two alternative methods for achieving circular polarization. Both approaches were ultimately unsuccessful, but they provided valuable insight that guided the final design decisions.

6.1 First Attempt: Corner-Truncated Single-Feed Method

The first approach employed a single-feed configuration with corner truncation (trim cut) to generate circular polarization. In this method, the perturbation introduced by the trimmed corners splits the fundamental mode into two orthogonal degenerate modes with the required 90° phase difference.



(a) Trimmed square ($L = W$)

Figure 14: Corner-truncated single-feed patch antenna (first unsuccessful attempt).

This design exhibited two significant limitations. First, the axial ratio bandwidth was extremely narrow, achieving acceptable circular polarization only at the center frequency of 5.3 GHz and degrading rapidly at the band edges. Second, and more importantly, I lacked a rigorous analytical model for determining the optimal truncation dimensions. Without a closed-form mathematical relationship between the trim size and the resulting axial ratio, optimization relied entirely on parametric sweeps of the trim variables—a time-consuming and inefficient process that ultimately failed to meet the broadband axial ratio requirement of < 3 dB across 5.1–5.5 GHz.

6.2 Second Attempt: Nearly-Square Patch Method

The second approach utilized a nearly-square patch geometry, wherein circular polarization is achieved by slightly detuning the patch dimensions ($W \neq L$) and carefully selecting the feed location to excite two orthogonal modes with the proper phase relationship.

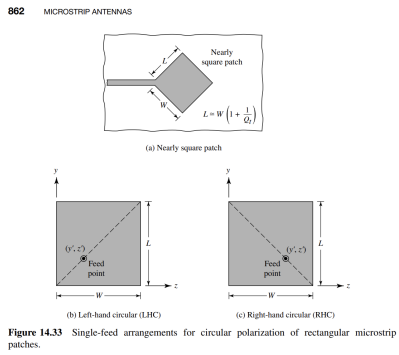


Figure 15: Nearly-square patch antenna with single feed (second unsuccessful attempt).

Although the analytical formulation for this method is well-established and my calculations were mathematically sound, the simulated axial ratio remained unacceptably high across the operating band. Furthermore, this approach proved to be highly sensitive to dimensional tolerances. Minor deviations from the calculated values—on the order of fractions of a millimeter—resulted in significant degradation of circular polarization performance. This extreme sensitivity renders the design impractical for fabrication, particularly when using foam board substrates where mechanical precision is limited.

6.3 Lessons Learned and Path Forward

The challenges encountered with the single-feed configurations highlighted the fundamental trade-off between design simplicity and performance robustness. While single-feed CP antennas offer the advantage of reduced feed network complexity, they inherently suffer from narrow axial ratio bandwidth and heightened sensitivity to manufacturing tolerances. These limitations motivated the adoption of the dual-feed square patch configuration presented in Section 3, which provides broader bandwidth, improved axial ratio stability, and greater resilience to fabrication inaccuracies at the cost of a slightly more complex feeding arrangement.